X (LQ input)

chroma swap

α^* -fuse

R G B

$\hat{Y}$ (network output)

luminance ✓ · chroma bias ×

$\hat{Y}_{\text{cp}} = X \odot \frac{Y_{\hat{Y}}}{Y_X}$

chroma ✓ (from X) · luminance ✓ (from $\hat{Y}$)

$\hat{Y}^*$ (fused)

MMSE-optimal per channel

Strop panel — Bayesian intuition as a probability-density schematic

$$p(\hat{Y}_{\text{cp}} \mid Y_c) \sim \mathcal{N}(Y_c, \sigma^2_{\text{LQ},c})$$

$$p(\hat{Y}^*_{\text{cp}} \mid Y_c) \sim \mathcal{N}(Y_c, \sigma^2_{\text{pred},c})$$

$$\alpha^*_c = \frac{\sigma^2_{\text{pred},c}}{\sigma^2_{\text{LQ},c} + \sigma^2_{\text{pred},c}}$$

$$\text{Var}(\hat{Y}^*_c) \le \min(\sigma^2_{\text{LQ},c}, \sigma^2_{\text{pred},c})$$

fused: variance ↓

Y_c

chroma value